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My main interest is to develop scientific methods and algorithms implemented in an efficient, intuitive and reliable data reduction and analysis software for the neutron imaging user community. I am particularly interested in the method development of advanced techniques such as Bragg edge and resonance imaging, grating interferometry, artificial intelligence (AI) assisted fast computed tomography (CT) reconstruction.

EXPERIENCE

JUNE 2020 – PRESENT

COMPUTER INSTRUMENT SCIENTIST, OAK RIDGE NATIONAL LABORATORY

- Lead neutron imaging software development at SNS and HFIR.
- Lead developer of cutting-edge Bragg edge analysis tool (Imaging Bragg Edge Analysis Tools for Engineering Structures - iBeatles).
- Lead developer of first world neutron instrument AI interface application.
- Lead developer of the internationally recognized neutron imaging Jupyter notebooks.
- Software contact to the imaging users to collect and develop software tools.
- Lead developer of the experiment specific Jupyter notebooks.
- Lead developer of user interfaces tools using python, PyQt, Marimo, Dash, etc.
- Co-developer of the web interface imaging tools (iNEUIT) used by users before coming to ORNL.
- Collect the team and users' software needs to either implement or create requirement documents.
- Provide documentation and step by step tutorial for all software tools implemented

JANUARY 2007 – JUNE 2020

COMPUTER SCIENTIST, OAK RIDGE NATIONAL LABORATORY

- Leader of neutron imaging software development at SNS and HFIR.
- Expert in developing user-friendly Python Jupyter notebooks for a broad and diverse user community at the HFIR CG-1D imaging beamline. These notebooks are capable of data processing, normalization, visualization and analysis.
- Created and maintains the neutron imaging tutorial web site (HTML, CSS) – https://neutronimaging.pages.ornl.gov; that contains both written and video tutorials of the software tools I develop.
- Leads Python training for neutron imaging user community (both on-site during beam time and remotely via conference calls).
- Point of contact for user-driven software requests.
- Actively collaborating with the user community in the development and implementation of advanced algorithms and Jupyter notebooks with user interface (UI).
- Developer of user-friendly interfaces in close collaboration with ORNL Neutron Scattering Division's instrument scientists using Python, MatLab, Java, and Vaadin.
- Leading the development and implementation of SANS reduction front end (Shaman)

Leading the development and implementation of all imaging front ends (iMARS for CG-1D, iBeatles for SNS imaging).

- Mentoring students and post-docs, teaching them Python, software development philosophy.
- Leading classes about imaging software and Python at domestic and international conferences/meetings.
- Actively presents software development at domestic and international conferences/meetings.
- Organizes workshops with international experts in imaging software.
- Lead point of contact for software development of Grating interferometry project LDRD.
- Co-investigator on a LDRD about extreme scale computing titled *Federating Scientific Instruments via Software Defined Infrastructure*

APRIL 2005 – JANUARY 2007

POST-DOCTORAL, OAK RIDGE NATIONAL LABORATORY

Developed 20+ tools (IDL, Java, Flex, perl) for all SNS beam lines to reduce, visualize and analyze their data.
Web-designed and created the neutron imaging web site using HTML, CSS, jquery, javascript.
Organized IDL training workshop.

JUNE 2003 – APRIL 2005

POST-DOCTORAL RESEARCH ASSOCIATE, OAK RIDGE NATIONAL LABORATORY

Led research on the study of production of short-lived radioactive species

JANUARY 1999 – MAY 2003

RESEARCH ASSISTANT, OAK RIDGE NATIONAL LABORATORY

Modeled, developed and manufactured target system to produce short lived radioactive species

EDUCATION

MAY 2003

PH.D. PHYSICS, UNIVERSITE DE VERSAILLES-SAINT-QUENTIN-EN-YVELINES,

DESIGN STUDIES OF HIGH-POWER FAST DIFFUSION-RELEASE TARGETS AND FAST VAPOR-TRANSPORT SYSTEMS FOR RADIOACTIVE ION BEAM

FRANCE

Specialized in technologies for large scale instrumentation

JUNE 1998

M.S. PHYSICS, UNIVERSITE DE VERSAILLES-SAINT-QUENTIN-EN-YVELINES, FRANCE

Minor in Mathematics

JUNE 1997

B.S. PHYSICS, UNIVERSITE DE VERSAILLES-SAINT-QUENTIN-EN-YVELINES, FRANCE

Minor in Mathematics

EXPERTISE

- Design and program of intuitive user interfaces
- Image reduction, processing and analysis for white beam and time-of-flight neutron imaging capabilities
- Image reconstruction and visualization (VG Studio)
- Python
- Dash, Marimo, PyQt

SYNERGISTIC ACTIVITIES

Committee member of the *International Society for Neutron Radiography* – September 2018 June 2024.

Member of SINE2020 (Science and Innovation with Neutrons in Europe in 2020) – 2017-present.

Reviewer for

- *Journal of Imaging*
- *Scientific Report*
- *Nuclear engineering and Design*
- *Journal of Neutron Research*
- *Quantum Beam Science*
- *Next Materials*
- *Inventions*

PUBLICATIONS (66)

1. *High spatial resolution neutron imaging of lithium-ion batteries: Correlating microstructure and lithium transport* – M. Gober, J. Amai, J. Torres, Y. Zhang, **J. Bilheux**, H. Z. Bilheux, G. Demeneghi, S. Tang, G. Smith and G. J. Nelson, 11-25, Journal of Power Sources 655, 237765.
2. *Neutron-friendly Li-ion battery coin cell for In Situ 3D visualization of Li plating* – M. Yusuf, J. M. LaManna, Y. Zhang, M. Preefer, B. Khaykovich, M. Dixit, C. Cao, D. P. Finegan, **J. Bilheux,** T. R. Tanim, E. J. Dufek, A. R. Dunlop, B. J. Polzin, A. N. Jansen, S. E. Trask, R. M. Kasse, D. N. Agyeman-Budu, P. P. Paul, J. N. Weker, M. F. Toney, 08 June 2025, Version 1, ChemRxiv
3. *Quantitative near-field water-air spray measurements at elevated pressures by neutron radiography imaging*, A. Clark, W. McCord, R. V. Ranganathan, Y. Zhang, **J. Bilheux**, Z. Zhang - Experiments in Fluids 65, 163 (2024).
4. *Operando Neutron Imaging-guided Gradient Design of Li-ion Solid Conductor for Extremely High Mass-loading Cathodes.* - H. Zhu, T. Ji, Y. Zhang, J. Torres, **J. Bilheux**, J. Wang, Nature Portfolio, (June 2024).
5. Li Dynamics in Mixed Ionic-Electronic Conducting Interlayer of All-Solid-State Li-metal Batteries. - D. Cao, Y. Zhang, T. Ji, X. Zhao, E. Cakmak, S. Ozcan, M. Geiwitz, **J. Bilheux**, K. Xu, Y. Wang, K. S. Burch, Q. Tu, H. Zhu, Nano Letters (January 2024).
6. *Visualization of in-situ chemical flow through sand using neutron radiography* - D. J. Morris, E. K. Morris, M. Nichols, J. D. Weidenhamer, H. Z. Bilheux, P. A. Cornwell, E. Stringfellow, Y. Zhang, **J. Bilheux**, Applied Radiation and Isotopes (December 2024).

7. *Jupyter Notebooks for Neutron Radiography Data Processing and Analysis*, **Jean-Christophe Bilheux**, Jiao Y. Lin and Hassina Z. Bilheux, Materials Research Proceedings 15 (2020) 198-204.
8. Brabazon, JW; Perfect, E; Gates, CH; Santodonato, LJ; Dhiman, I; Bilheux, HZ; **Bilheux, J-C**; Mckay, LD - *Spontaneous Imbibition of a Wetting Fluid Into a Fracture with Opposing Fractal Surfaces: Theory and Experimental Validation*, Fractals, 27, 1, 1940001, 2019, World Scientific Publishing Company
9. Ossler, Frederik; Santodonato, Louis J; Warren, Jeffrey M; Finney, Charles EA; **Bilheux, Jean-Christophe**; Mills, Rebecca A; Skorpenske, Harley D; Bilheux, Hassina Z - *In situ monitoring of hydrogen loss during pyrolysis of wood by neutron imaging*, Proceedings of the Combustion Institute, 37, 2, 1273-1280, 2019, Elsevier
10. Wang, Tianhao; Jiang, Chenyang; Bilheux, Hassina; Dhiman, Indu*; Bilheux, Jean-Christophe*; Crow, Morris; McDonald, Landen; Robertson, Lee; Kardjilov, Nikolay; Pynn, Roger - *New Polarized Neutron Imaging Capability at Oak Ridge National Laboratory*, APS Meeting Abstracts,2019
11. Nie, Ziyang; McCormack, Patrick; Bilheux, Hassina Z; **Bilheux, Jean C**; Robinson, J Pierce; Nanda, Jagjit; Koenig Jr, Gary M - *Probing lithiation and delithiation of thick sintered lithium-ion battery electrodes with neutron imaging*, Journal of Power Sources, 419, 127-136, 2019, Elsevier
12. Wang, T; Jiang, CY; Bilheux, HZ; Dhiman, I; **Bilheux, JC**; Crow, L; McDonald, L; Robertson, L; Kardjilov, N; Pynn, R - *Improving polarized neutron imaging for visualization of the Meissner effect in superconductors*, Review of Scientific Instruments, 90, 3, 33705, 2019, AIP Publishing
13. Nie, Ziyang; McCormack, Patrick; Bilheux, Hassina; **Bilheux, Jean**; Robinson, Pierce; Koenig, Gary; - *Fabrication and Characterization of Thick Sintered Lithium-Ion Battery Full Cells*, Meeting Abstracts, 5, 519-519, 2019, The Electrochemical Society
14. **Bilheux, Jean-Christophe**; Bilheux, Hassina; Lin, Jiao; Lumsden, Ian; Zhang, Yuxuan - *Neutron imaging analysis using jupyter Python notebook*, Journal of Physics Communications, 3, 8, 83001, 2019, IOP Publishing
15. Nie, Ziyang; McCormack, Patrick; Bilheux, Hassina; **Bilheux, Jean**; Robinson, James; Koenig, Gary; - *Characterization of lithiation/delithiation of thick sintered lithium-ion battery electrodes using neutron imaging*, ABSTRACTS OF PAPERS OF THE AMERICAN CHEMICAL SOCIETY,257, 2019, AMER CHEMICAL SOC 1155 16TH ST, NW, WASHINGTON, DC 20036 USA
16. Zhang, Yuxuan; **Bilheux, Jean**; Bilheux, Z Hassina; Lin, Jiao - *An Interactive Web-based Tool to Guide the Preparation of Neutron Imaging Experiments at Oak Ridge National Laboratory*, Journal of Physics Communications, 2019, IOP Publishing
17. Herringer, Susan N; Ryzewski, Krysta; Bilheux, Hassina Z; Bilheux, Jean-Christophe; Sheldon, Brian W - *Evaluation of segregation in Roman sestertius coins,Journal of materials science*, 53, 3, 2161-2170, 2018, Springer US
18. Xie, Qingge; Song, Gian; Gorti, S; Stoica, Alexandru Dan; Radhakrishnan, Balasubramaniam; *Bilheux, Jean-Christophe*; Kirka, M; Dehoff, R; Bilheux, Hassina Z; An, Ke - *Applying neutron transmission physics and 3D statistical full-field model to understand 2D Bragg-edge imaging*, Journal of Applied Physics, 123, 7, 74901,2018, AIP Publishing
19. *Bilheux, Jean - NeuNorm: Neutron Imaging Normalization Library.*, J. Open Source Software, 3, 29, 815, 2018,
20. Myhre, Kristian G; Zhang, Yuxuan; Bilheux, Hassina Z; Johnson, Jared A; **Bilheux, Jean-Christophe**; Miskowiec, Andrew J; Hunt, Rodney Dale - *Nondestructive Tomographic Mapping of Uranium and Gadolinium Using Energy-Resolved Neutron Imaging*, 2018, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States)
21. Zhang, Yuxuan; Myhre, Kristian G; Bilheux, Hassina Z; Tremsin, Anton S; Johnson, Jared A; **Bilheux, Jean-Christophe**; Miskowiec, Andrew J; Hunt, Rodney Dale; Santodonato, Louis J; Molaison, Jamie J - *Neutron Resonance Radiography and Application to Nuclear Fuel Materials*, 2018, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States)
22. Zhang, Y; Chandran, KS Ravi; Jagannathan, M; Bilheux, HZ; **Bilheux, JC** - *The Nature of Electrochemical Delithiation of Li-Mg Alloy Electrodes: Neutron Computed Tomography and*

Analytical Modeling of Li Diffusion and Delithiation Phenomenon, Journal of The Electrochemical Society, 164, 2, A28-A38, 2017, The Electrochemical Society
23. Herringer, Susan N; Ryzewski, Krysta; Bilheux, Hassina Z; **Bilheux, Jean-Christophe**; Sheldon, Brian W - The Application of Neutron imaging to investigate the Design and Manufacture of a Bronze Oinochoe,Restaurierung und Archäologie, 8, 35-43, 2017,
24. Zhang, Yuxuan; **Bilheux, Jean-Christophe** - *ImagingReso: A Tool for Neutron Resonance Imaging*, Journal of Open Source Software, 2, 19, 2017, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States)
25. Song, Gian; Lin, Jiao; **Bilheux, Jean**; Xie, Qingge; Santodonato, Louis; Molaison, Jamie; Skorpenske, Harley; M Dos Santos, Antonio; Tulk, Chris; An, Ke; *Characterization of Crystallographic Structures Using Bragg-Edge Neutron Imaging at the Spallation Neutron Source*, Journal of Imaging, 3, 4, 65, 2017, Multidisciplinary Digital Publishing Institute
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27. Perfect, Edmund; Gates, Christopher Hallett; Brabazon, Jared William; Santodonato, Louis Joseph; Dhiman, Indu; Bilheux, Hassina; **Bilheux, Jean-Christophe**; Lokitz, Bradley S - *Complex Contact Angles Calculated from Capillary Rise Measurements on Rock Fracture Faces*, AGU Fall Meeting Abstracts, 2017,
28. Adlmann, Franz A; Pálsson, Gunnar Karl; **Bilheux, JC**; Ankner, JF; Gutfreund, P; Kawecki, M; Wolff, Max - *Överlåtaren: a fast way to transfer and orthogonalize two-dimensional off-specular reflectivity data*, Journal of Applied Crystallography, 49, 6, 2091-2099, 2016, International Union of Crystallography
29. Bilheux, Hassina Z; Song, Gian; An, Ke; **Bilheux, Jean-Christophe**; Kirka, Michael M; Dehoff, Ryan R; Santodonato, Louis J; Gorti, Sarma B; Radhakrishnan, Balasubramaniam; Xie, Qingge - *Advances in Neutron Radiography: Application to Additive Manufacturing Inconel 718*, 2016, Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States).
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33. Santodonato, Lou; Bilheux, Hassina; Bailey, Barton; **Bilheux, Jean**; Nguyen, Phong; Tremsin, Anton; Selby, Doug; Walker, Lakeisha - *The CG-1D neutron imaging beamline at the oak Ridge National Laboratory high flux isotope reactor*, Physics Procedia, 69, 104-108, 2015, Elsevier
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35. Bilheux, Hassina; **Bilheux, Jean**; Tremsin, Anton; Santodonato, Lou; Dehoff, Ryan; Kirka, Mike; Bailey, Barton; Keener, Scott; Herwig, Ken - *Overview of the neutron radiography and computed tomography at the oak ridge national Laboratory and Applications*, ASNT 24th Research Symposium 2015, 18-23, 2015
36. Arnold, Owen; **Bilheux, Jean-Christophe**; Borreguero, JM; Buts, Alex; Campbell, Stuart I; Chapon, L; Doucet, Mathieu; Draper, N; Leal, R Ferraz; Gigg, MA - *Mantid—Data analysis and visualization package for neutron scattering and μ SR experiments*, Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, 764, 156-166, 2014, North-Holland

37. Lance, Michael J; Bilheux, Hassina; **Bilheux, Jean-Christophe**; Voisin, Sophie; Sluder, C Scott; Stevenson, Joseph - *Neutron Tomography of Exhaust Gas Recirculation Cooler Deposits*, 2014, SAE Technical Paper
38. Alton, Gerald D; **Bilheux, J-C**; Zhang, Y; Liu, Yuan - *Molecular-flow time properties of RIB type vapor-transport systems using a fast-valve*, Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms, 329, 30-39, 2014
39. Ryzewski, Krysta; Bilheux, Hassina Z; Herringer, Susan N; **Bilheux, Jean-Christophe**; Walker, Lakeisha; Sheldon, Brian W - *The Use and Refinement of Neutron Imaging Techniques for Archaeological Artifacts*, Advances in Archaeological Practice, 2, 2, 91-103, 2014,Cambridge University Press
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41. Cekanova, Maria; Donnell, Robert; Bilheux, Hassina; **Bilheux, Jean-Ch** - *Neutron imaging: Detection of cancer using animal model*, Proceedings of the 2014 Biomedical Sciences and Engineering Conference, 1-4, 2014, IEEE
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43. Bilheux, H; Crawford, K; Walker, L; Voisin, S; Kang, M; Harvey, M; Bailey, B; Phillips, M; *Bilheux, JC*; Berry, K - *Neutron imaging at the Oak Ridge National Laboratory: present and future capabilities*, Physics Procedia, 2013,
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45. Ryzewski, Krysta; Herringer, Susan; Bilheux, Hassina Z; Walker, Lakeisha MH; Sheldon, Brian; Voisin, Sophie; **Bilheux, Jean-Christophe**; Finocchiaro, Vincenzo - Neutron Imaging of Archaeological Bronzes, 2013, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States); High Flux.
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